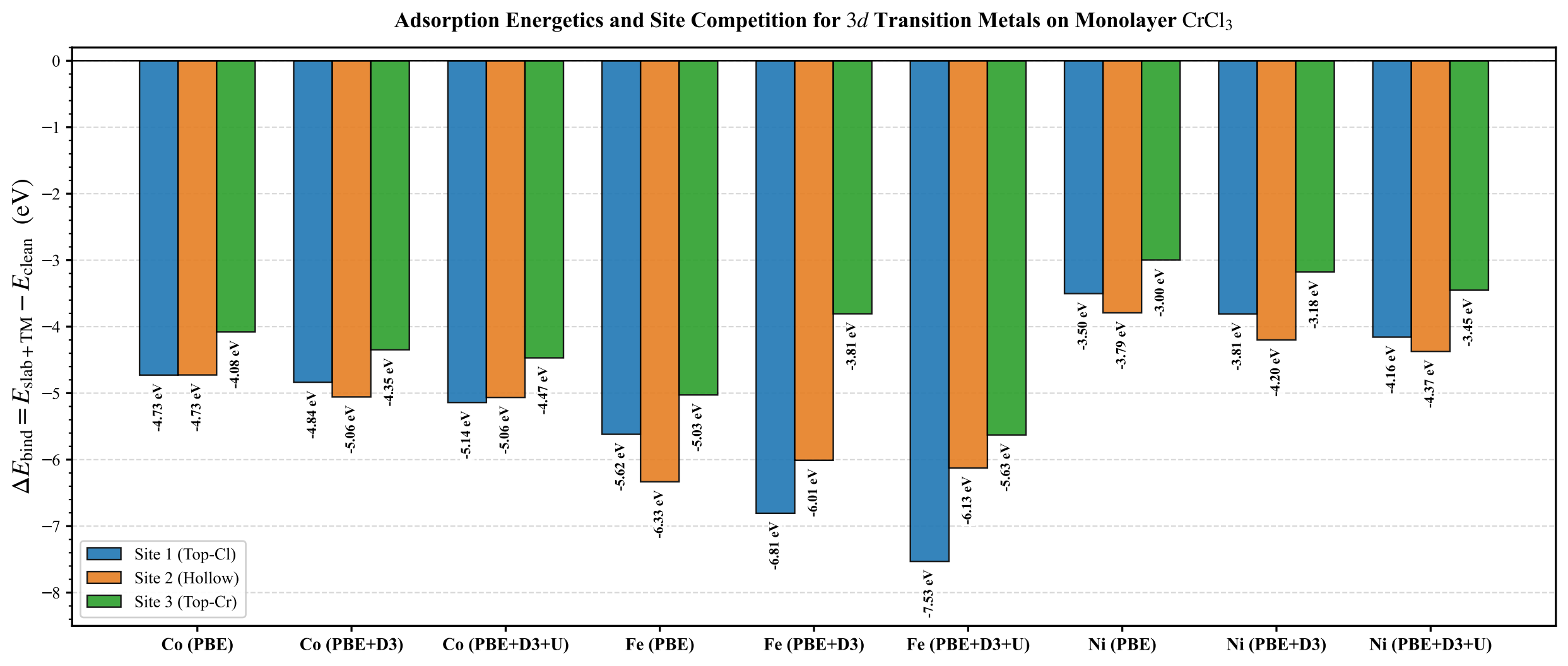
Adsorption Energetics and Site Competition for 3d Transition Metals on Monolayer CrCl3
ΔEbind = Eslab + TM − Eclean (eV)
0
−1
−2
−3
−4
−5
−6
−7
−8
-4.73 eV
-4.73 eV
-4.08 eV
-4.84 eV
-5.06 eV
-4.35 eV
-5.14 eV
-5.06 eV
-4.47 eV
-5.62 eV
-6.33 eV
-5.03 eV
-6.81 eV
-6.01 eV
-3.81 eV
-7.53 eV
-6.13 eV
-5.63 eV
-3.50 eV
-3.79 eV
-3.00 eV
-3.81 eV
-4.20 eV
-3.18 eV
-4.16 eV
-4.37 eV
-3.45 eV
Site 1 (Top-Cl)
Site 2 (Hollow)
Site 3 (Top-Cr)
Co (PBE)
Co (PBE+D3)
Co (PBE+D3+U)
Fe (PBE)
Fe (PBE+D3)
Fe (PBE+D3+U)
Ni (PBE)
Ni (PBE+D3)
Ni (PBE+D3+U)